

SpacECE Internship Assessment

Sales Data Analysis & Forecasting

Project Report

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This report presents the complete project workflow, code methodology, analytical charts, model output, key insights, recommendations, and repository details.

GitHub Repository
github.com/Kunallhundare/SpacECE

1. Project Description

The project analyzes a sales dataset containing 9,800 records and originally 18 columns. The work follows an end-to-end data analytics workflow: data inspection, data cleaning, date conversion, feature engineering, exploratory data analysis (EDA), statistical summary, and monthly sales forecasting using Linear Regression.

The objective is to transform raw sales data into useful business insights and identify sales patterns across time, regions, product categories, customer segments, and individual products.

Project Workflow

Data Collection → Data Understanding → Data Cleaning → Feature Engineering → EDA → Statistical Analysis → Sales Forecasting → Business Recommendations

2. Dataset & Preprocessing

Dataset item	Result
Rows	9,800
Columns after feature engineering	22
Original data columns	18
Missing Postal Codes	Filled using median
Duplicate rows checked	Yes
Date columns	Converted to datetime

Code Methodology

Missing values: The Postal Code column was completed using its median value.

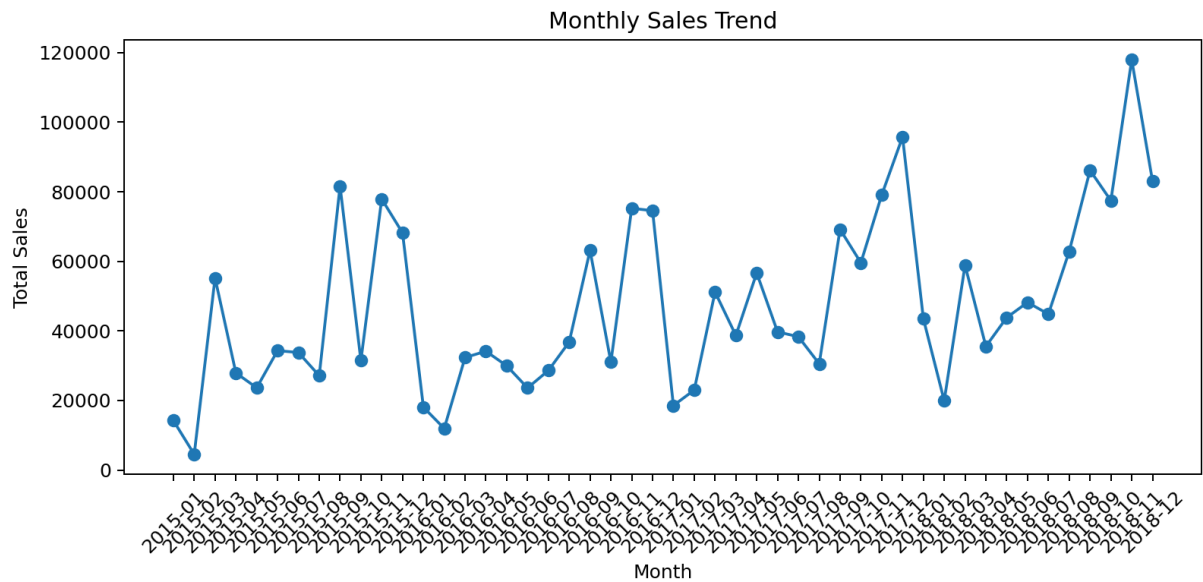
Dates: Order Date and Ship Date were converted to datetime format.

New features: Shipping Days, Year, Month_Number, and Month were created to support time-based analysis.

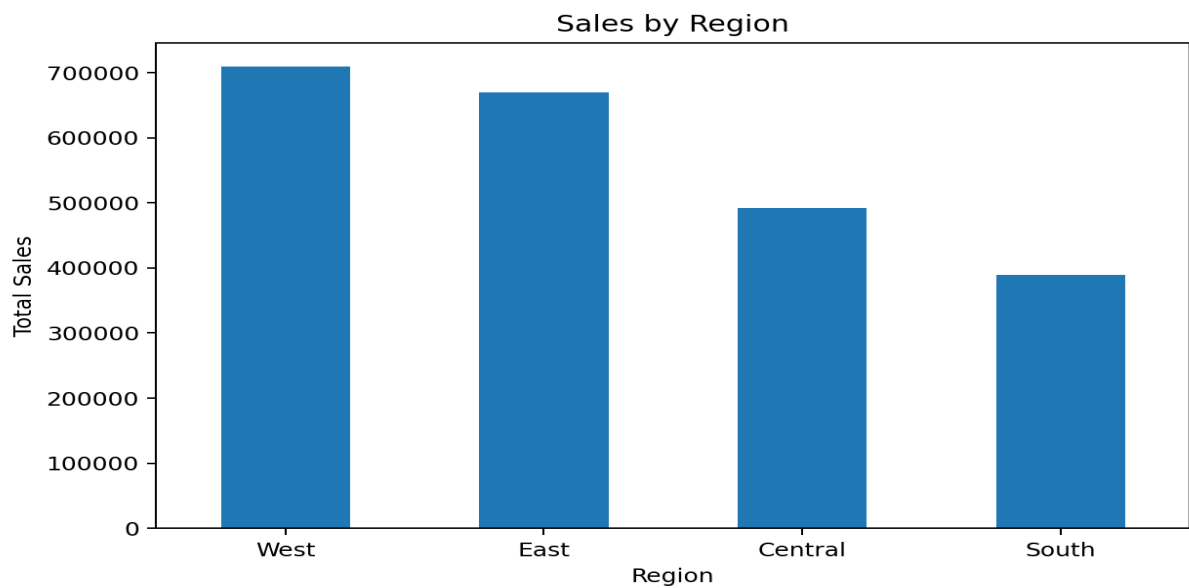
3. Charts & Output

The following charts were generated from the project dataset and reflect the analysis implemented in the Jupyter Notebook.

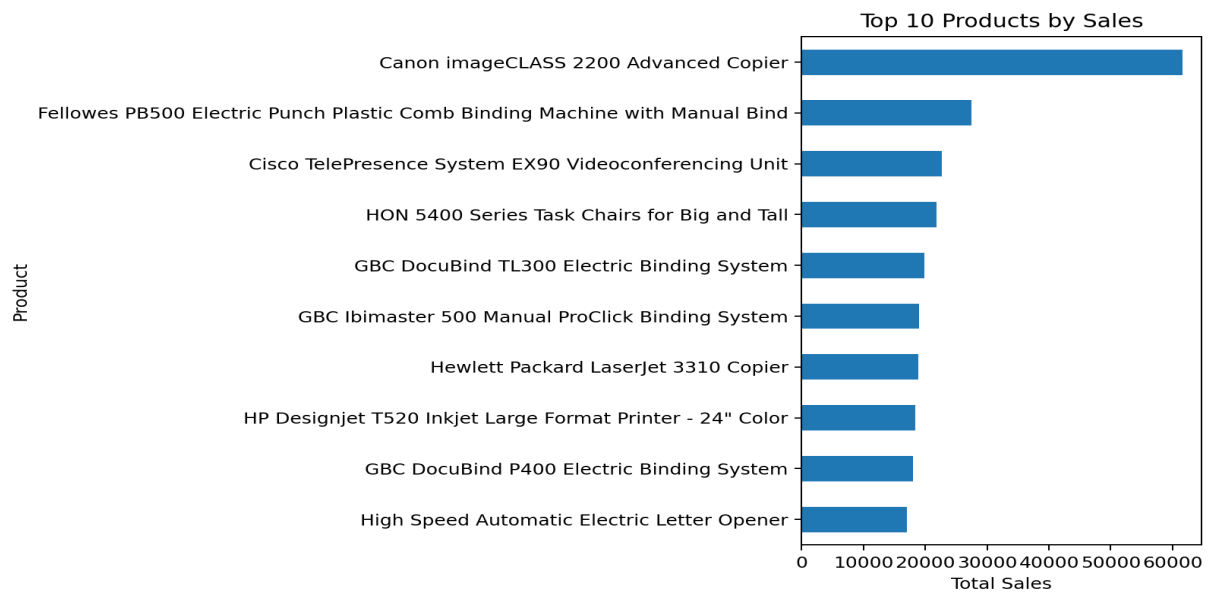
Monthly Sales Trend



Sales by Region



Top 10 Products by Sales



4. Forecasting Model Output

The notebook uses a Linear Regression model for monthly sales forecasting. The first 80% of monthly observations are used for training and the final 20% for testing. Time index and month-based seasonal indicators are used as input features.

Metric	Result
Training months	38
Testing months	10
R-Squared (R^2)	0.637
RMSE	14,478.82
MAE	10,520.28

Model Interpretation

The model achieved an R^2 score of approximately 0.637, meaning it explains a meaningful portion of the variation in monthly sales but does not capture every change. The RMSE and MAE indicate that prediction errors remain, so the model is suitable as a baseline forecasting approach and can be improved with additional features or advanced forecasting methods.

5. Key Insights

- **Sales distribution is highly skewed:** Average sales are 230.77, while median sales are only 54.49. A smaller number of very large orders increase the average.
- **Technology is the top category:** Technology generated the highest category sales at 827,455.87.
- **West is the leading region:** The West recorded the highest regional sales at 710,219.68.
- **Seasonal sales pattern exists:** The highest monthly sales value in the dataset occurred in 2018-11 at 117,938.15.
- **Forecasting has moderate baseline performance:** The Linear Regression model achieved $R^2 = 0.637$, showing useful but improvable predictive performance.

6. Recommendations

- **Prioritize Technology products:** Maintain sufficient inventory and strengthen promotions for high-performing Technology products.
- **Strengthen the West region:** Continue successful marketing and sales strategies in the highest-performing region while applying similar learnings to lower-performing regions.
- **Prepare for high-demand periods:** Use monthly trends to plan inventory, staffing, and campaigns before stronger sales periods.
- **Improve forecasting:** Add more historical data and experiment with advanced models and additional business features to reduce prediction error.
- **Monitor high-value orders separately:** Because sales are skewed, analyze large transactions independently so they do not hide typical customer purchasing behavior.

7. GitHub Repository Link

Repository: github.com/Kunallhundare/SpaceCE

The repository contains the Jupyter Notebook, dataset, and project documentation.

Conclusion

This project demonstrates practical skills in Python-based data analysis, data cleaning, feature engineering, visualization, statistical interpretation, and basic machine learning forecasting. The analysis converts a raw sales dataset into structured findings that can support business planning and future improvements.